TRACE LEARN

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Project ID · Version:
F26-186 · fb94119

Iteration 1: C++ submission → tests → diagnosis → hint

Approved Problem Statement:
Students repeat the same misconception across problems, but platforms judge each submission on its own.

PROPOSAL ACTIONS

Panel Comments	Changes made
Use an LLM for user experience	1. LLM integration will be used to help teachers grade student submissions. 2. LLM used to generate hints for the students from the evidences

ITERATION PLAN

Element	Status	Evidence	Owner
RBAC	✓	JWT roles, be7897b, fc16065	Eman, Umaima
Monaco integration	✓	Editor in C++ mode; 5619739	Eman
C++ execution pipeline	✓	g++ in a bubblewrap sandbox, 8ef0efb, c30e5aa	Eman, Umaima
Schema normalisation	✓	21 tables, unique attempt numbers; 1834765, fb94119	Rayaan
Draft misconception taxonomy	✓	3 Python and 10 C++ codes; 12c3c26	Rayaan, Umaima
Evidence extraction pipeline	✓	5 evidence families; b88a962, 12c3c26	Rayaan, Umaima
Submission logging	✓	b88a962	Umaima
Testing	✓	191/191 tests pass; CI passing; 40/40 correct solutions give no misconception- fb94119	Eman (QA), Umaima (CI)
Deployment	✓	https://trace-learn.vercel.app/	Rayaan

BLOCKERS & RECOVERY

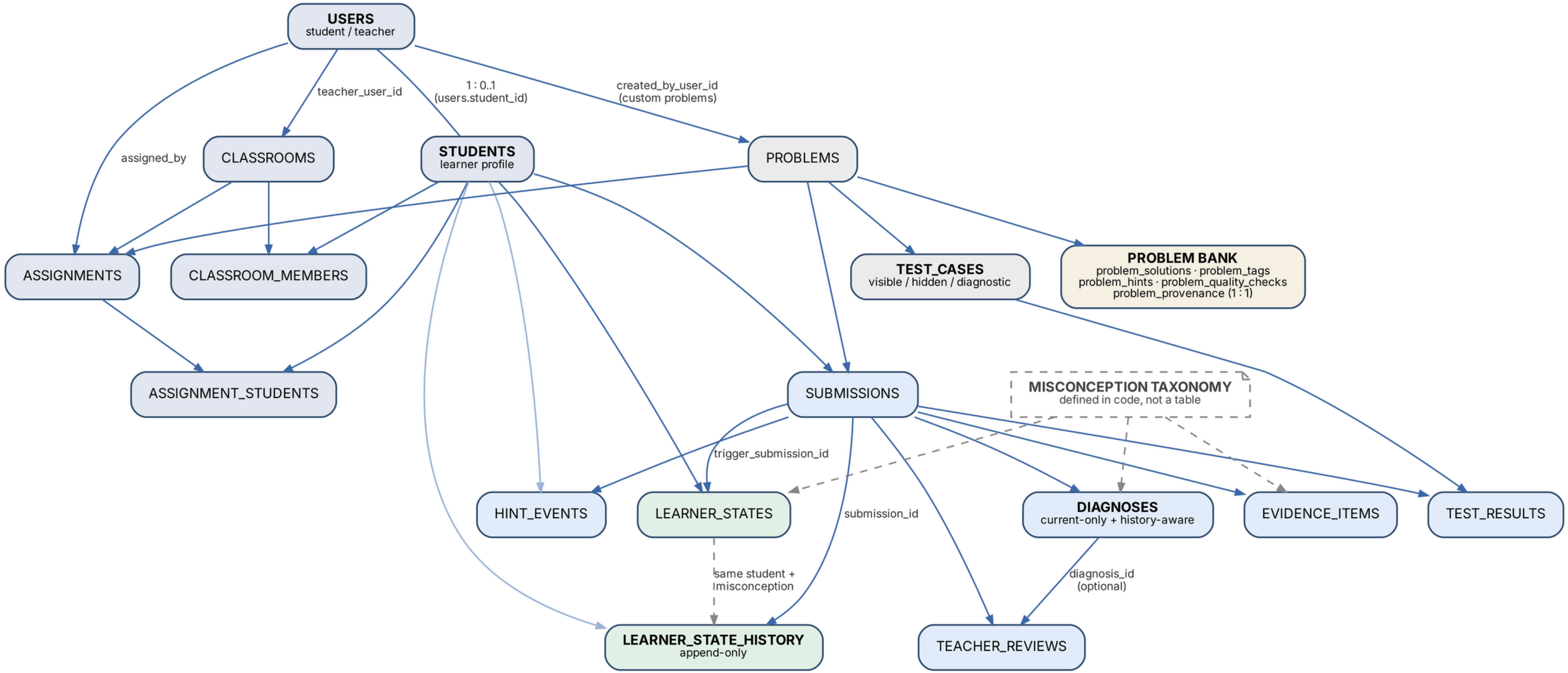
Blocker	Recovery	Date
C++ submissions did not run on Windows	Gave the compiled program its full path (fb94119)	Oct 1, 2026
C++ evidence was not scored; correct code was flagged for review	Mapped the C++ codes into scoring (12c3c26); added a “no misconception” result when all tests pass (fb94119)	Sep 30, 2026
Two submissions sent at once could get the same attempt number	Added a unique constraint with a retry (fb94119)	Oct 8, 2026

REQUIREMENTS & ACCEPTANCE

Sign in → open assigned problem → submit C++ → tests → diagnosis + hint → teacher review

Requirement	Acceptance check → result
FR-05 C++ run with limits	Correct code → success code that does not compile → syntax_error infinite loop → timeout at 2 s → **Pass**
FR-06 Hidden tests stay hidden	Students never receive hidden or diagnostic input or output → **Pass** (77/77 API checks)
NFR-02 Isolation	Student code cannot read host files, use the network or read env vars (Linux + bubblewrap) → **Pass**

DESIGN EVOLUTION



CORE MODULE AND VALIDATION

Input: C++ source + problem ID, sent from the Monaco editor

Processing: g++ in a bubblewrap sandbox → 9 tests (2 s, 256 MB) → evidence → 2 diagnoses → hint

Output: status, test results, diagnosis, hint level 1–3. Needs g++, bubblewrap, FastAPI, SQLite

Tests: 191/191 pass. 40/40 correct solutions → no misconception. 40/43 variants labelled

Limit: history-aware mode does not change a prediction yet

REPO, OWNERS & NEXT

github.com/Rayaan-Raza/TraceLearn - v0.2.0-mid1

Member	Ownership
Rayaan	POC core (evidence, diagnosis, learner state); the 40-problem C++ bank; deployment
Umaima	Bubblewrap sandbox; auth hardening and security tests; UI–API integration; submission logging; CI/CD
Eman	First RBAC; Monaco in C++ mode; the C++ compile and run pipeline; Iteration 1 QA

GenAI Usage:

Claude (L3)
Cursor (L2)
*all reviewed/verified

external tools/datsets

g++, bubblewrap, Monaco
CodeNet, PROGpedia and CodeStream
(licences checked)

NEXT ITERATION

- Problem & test bank expansion
- Annotation handbook & labelling protocol
- Learner-state modelling
- Misconception classifier XGBoost model
- Adaptive hint framework (initial levels- LLM introduced)
- Submission view